

# DHRUV PATEL

dhrvptl.dev@gmail.com | (732) 692-7041 | linkedin.com/in/dhruv-patel-dev23 | github.com/dhrv-ptl | U.S. Citizen

## EDUCATION

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### Rutgers University

Sep 2022 – May 2026

B.S. Computer Science & Data Science (GPA: 3.61/4.00 | Dean's List: 6/8 Semesters)

New Brunswick, NJ

- **Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Software Methodology, Computer Architecture, Data Management, Machine Learning, Databases.

## WORK EXPERIENCE

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### Backend Software Developer Intern

Sep 2025 – Jun 2026

Medidata Solutions (Dassault Systèmes)

New York, NY

- Built Java backend services and automated validation workflows for internal data pipelines, reducing manual regression tracking by **30%** through reusable business rules, stronger input checks, and regression-status reporting.
- Owned relational data models, permission rules, validation logic, and workflow configurations for **1,000+** production users, improving data integrity, access control, and reliability across internal enterprise systems.
- Triaged production data and integration issues by tracing execution logs, isolating schema-validation edge cases, and documenting recurring failure patterns to improve debugging speed and support handoffs.
- Improved deployment readiness by building automated test harnesses, validating fixes across dev/QA/prod, and supporting Git/Jenkins CI/CD releases with clearer rollback and verification steps.

### Platform Software Developer Intern

May 2025 – Aug 2025

Medidata Solutions (Dassault Systèmes)

New York, NY

- Consolidated **74+** legacy workflow scripts into **38** modular automation workflows, removing duplicate execution paths and making recurring jobs easier to test, monitor, and maintain.
- Implemented Java service-layer logic and unit/integration tests for internal data-ingestion workflows, raising coverage above **80%** and reducing release risk around validation and transformation changes.
- Built Jira API ingestion tooling to aggregate sprint metadata, generate tickets, and surface delivery metrics for product and engineering stakeholders during sprint planning and release tracking.
- Validated releases by reproducing race conditions, verifying runtime fixes, and supporting deployments across dev, QA, and production environments to catch workflow issues before release.

### Software Engineer (Contract)

May 2025 – Nov 2025

Tingle for Mayor 2025 Campaign

Edison, NJ

- Designed a Node.js/Express REST API for donation tracking and secure form submissions, supporting **15K+** monthly visitors with structured request validation and reliable payment-state handling.
- Integrated Apple Pay and Google Pay checkout flows that processed **\$75K+** in digital contributions while maintaining sub-2-second page loads through lightweight endpoints and optimized client-side form logic.
- Increased form-completion conversion by **40%** by analyzing telemetry, identifying validation drop-offs, and improving payment-form data handling for cleaner user input and fewer failed submissions.

## TECHNICAL SKILLS

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**Languages:** Python, Java, SQL, TypeScript, JavaScript, Bash, C++, Go

**Infrastructure:** AWS, Terraform, CloudFormation, Docker, Kubernetes, EKS, ECS, Linux, CI/CD, Jenkins, Git, CloudWatch, Splunk, Datadog

**Cloud/Networking:** VPCs, Subnets, Route Tables, Security Groups, NAT Gateways, ALB/ELB, Route53, API Gateway, AWS WAF, AWS Network Firewall

**Backend/Data:** REST APIs, Microservices, PostgreSQL, MySQL, MongoDB, ETL/ELT, Data Modeling, Pandas, NumPy, Jira APIs

## PROJECTS

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### LineWise: NYC Transit Delay Alert API

Mar 2026 – Apr 2026

- Built a Python/FastAPI delay-alert service using NYC transit feed data and scheduled route metadata, containerizing the API with Docker and exposing route/station risk predictions through REST endpoints.
- Deployed the service with Terraform-managed AWS infrastructure, including VPC subnets, route tables, security groups, NAT gateways, ALB routing, Route53 records, API Gateway access, and AWS WAF-style request filtering.
- Added Kubernetes manifests for local orchestration testing and CloudWatch/Splunk-style observability for health checks, latency, 5xx errors, and failed upstream feed refreshes.

### Medipix: AI Photo Management System

May 2025 – Aug 2025

- Developed a React/AWS cloud-backed image ingestion and semantic search system with Okta authentication, AWS Lambda processing, and Google Drive integration for searchable photo assets.
- Reduced manual metadata tagging by **80%** by integrating facial-recognition clustering and background indexing jobs that generated searchable labels, grouped similar images, and improved retrieval speed.

### IronTrack: Fitness Analytics Platform

Jan 2026 – Mar 2026

- Built a React/TypeScript workout analytics platform for logging training metrics, aggregating weekly trends, and tracking PR and volume progression from structured Supabase records.